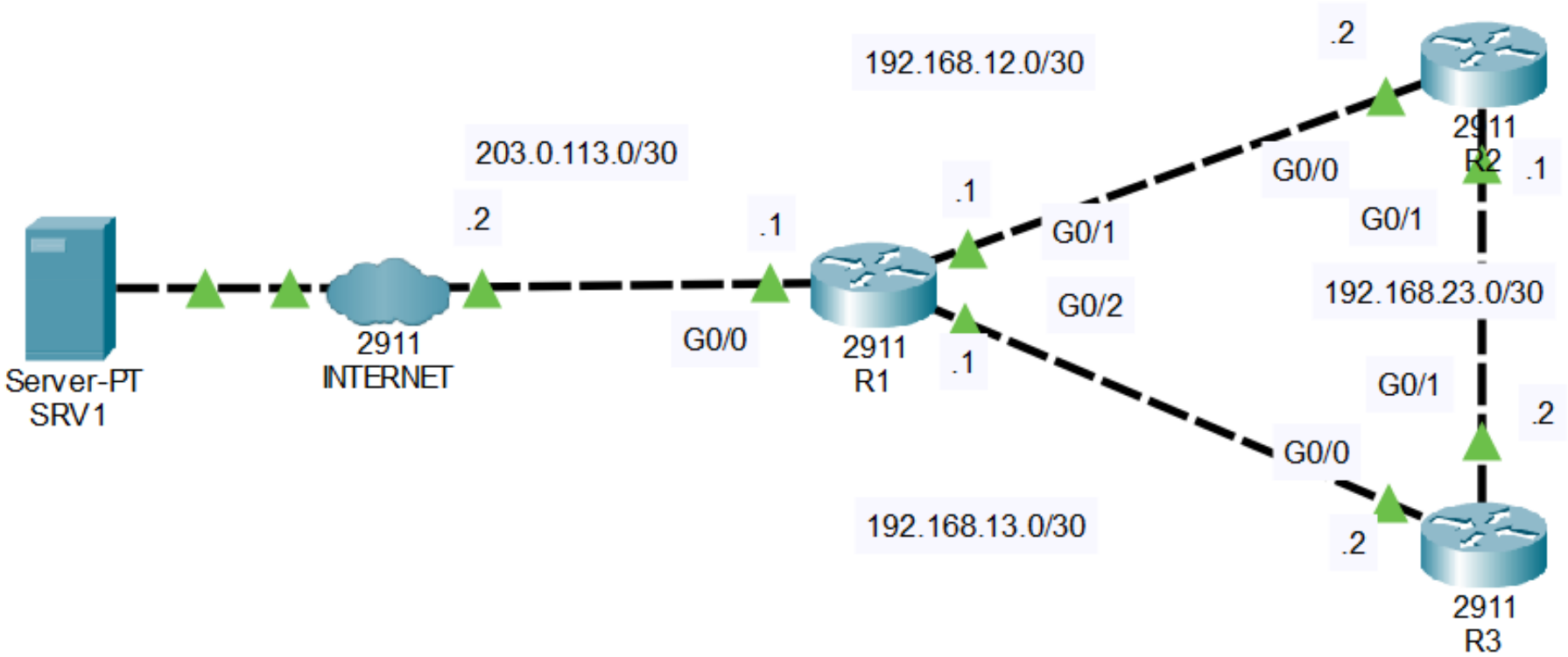


ROUTING HAS BEEN PRECONFIGURED
(default route on R1, OSPF on all routers w ith 'netw ork 0.0.0.0 255.255.255.255 area 0')

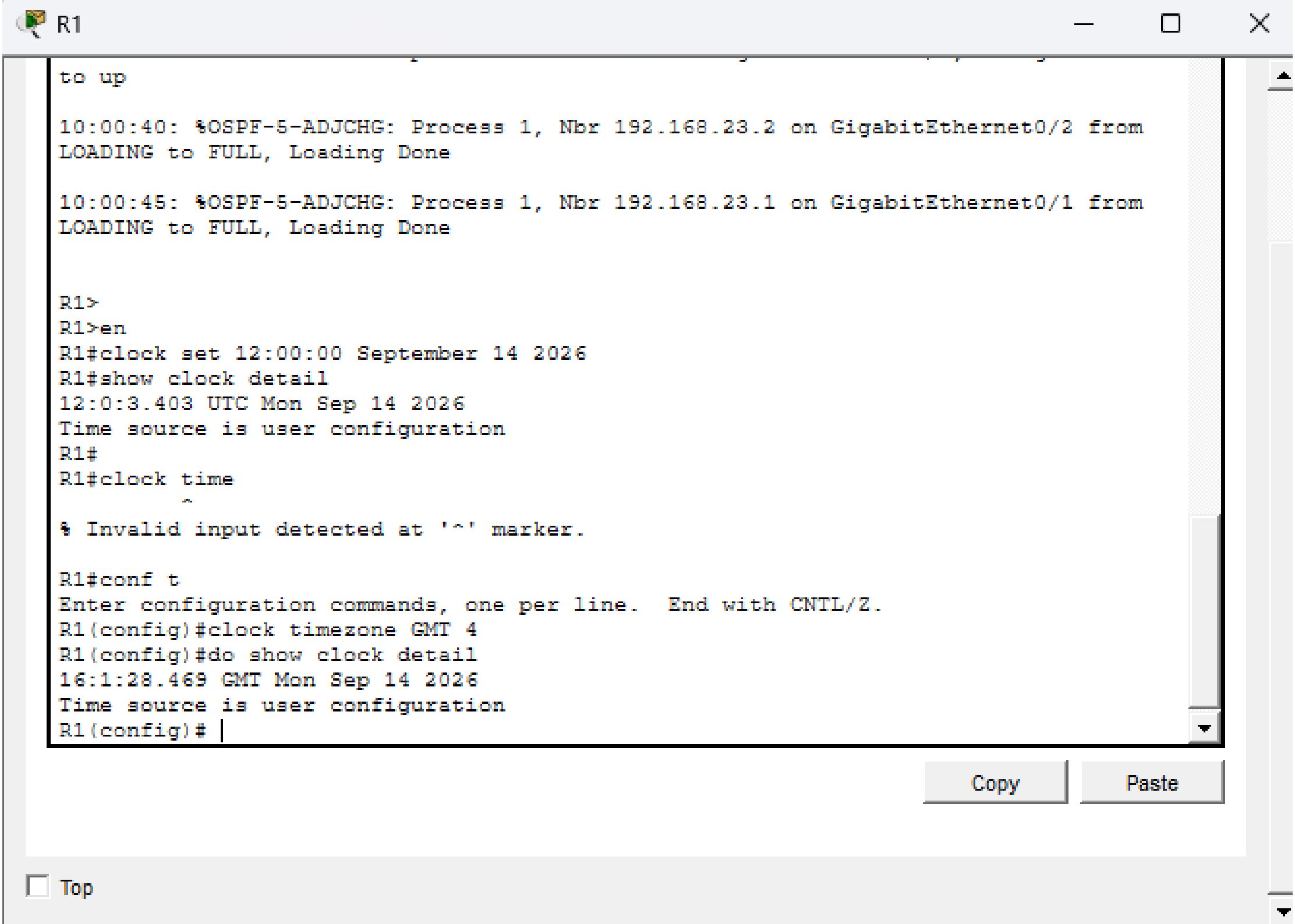
1. Configure the softw are clock on R1, R2, and R3 to 12:00:00 September 14 2026 (UTC).

2. Configure the time zone of R1, R2, and R3 to match your ow n.

3. Configure R1 to synchronize to NTP server 1.1.1.1 over the Internet.
   What stratum is 1.1.1.1? What stratum is R1?

4. Configure R1 as a stratum 8 NTP master.
   Synchronize R2 and R3 to R1 w ith authentication.
   *the 'ntp source' command is not available in Packet Tracer, so just use the physical interface IP addresses of R1.

5. Configure NTP to update the hardw are calendars of R1, R2, and R3.
   *you can't view the calendar in Packet Tracer


R1
to up
10:00:40: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.23.2 on GigabitEthernet0/2 from LOADING to FULL, Loading Done
10:00:45: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.23.1 on GigabitEthernet0/1 from LOADING to FULL, Loading Done
R1>
R1>en
R1#clock set 12:00:00 September 14 2026
R1#show clock detail
12:0:3.403 UTC Mon Sep 14 2026
Time source is user configuration
R1#
R1#clock time
^
% Invalid input detected at '^' marker.
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#clock timezone GMT 4
R1(config)#do show clock detail
16:1:28.469 GMT Mon Sep 14 2026
Time source is user configuration
R1(config)#
Copy
Paste
Top

R2

```
Press RETURN to get started!


%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state
to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state
to up

10:00:40: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.23.2 on GigabitEthernet0/1 from
LOADING to FULL, Loading Done

10:00:45: %OSPF-5-ADJCHG: Process 1, Nbr 203.0.113.1 on GigabitEthernet0/0 from
LOADING to FULL, Loading Done


R2>
R2>
R2>en
R2#clock set 12:00:00 September 14 2026
R2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#clock timezone gmt 4
R2(config)#do show clock detail
16:0:16.620 gmt Mon Sep 14 2026
Time source is user configuration
R2(config)#
```

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R3

```
Press RETURN to get started!


%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state
to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state
to up

10:00:40: %OSPF-5-ADJCHG: Process 1, Nbr 203.0.113.1 on GigabitEthernet0/0 from
LOADING to FULL, Loading Done

10:00:40: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.23.1 on GigabitEthernet0/1 from
LOADING to FULL, Loading Done


R3>
R3>
R3>en
R3#clock set 12:00:00 september 14 2026
R3#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#clock timezone GMT 4
R3(config)#do show clock detail
16:0:12.616 GMT Mon Sep 14 2026
Time source is user configuration
R3(config)#
```

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Top

```
R1(config)#
R1(config)#
R1(config)#
R1(config)#
R1(config)#
R1(config)#
R1(config)#ntp server 1.1.1.1
R1(config)#do sh ntp association

address         ref clock       st   when     poll    reach  delay          offset            disp
 ~1.1.1.1       .INIT.          16   -        64      0      0.00           0.00              0.01
 * sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured
R1(config)#









R1 con0 is now available



Press RETURN to get started.

R1>
R1>en
R1#show ntp association

address         ref clock       st   when     poll    reach  delay          offset            disp
*~1.1.1.1       127.127.1.1     1    33       64      377    0.00           0.00              0.48
 * sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured
R1#
```

```
R1#
R1#
R1#show ntp status
Clock is synchronized, stratum 2, reference is 1.1.1.1
nominal freq is 250.0000 Hz, actual freq is 249.9990 Hz, precision is 2**24
reference time is E3695EA8.00000241 (22:52:24.577 UTC Tue Dec 29 2020)
clock offset is 0.00 msec, root delay is 1.00  msec
root dispersion is 56.97 msec, peer dispersion is 0.48 msec.
loopfilter state is 'CTRL' (Normal Controlled Loop), drift is - 0.000001193 s/s
system poll interval is 6, last update was 14 sec ago.
R1#show clock detail
2:52:46.320 GMT Wed Dec 30 2020
Time source is NTP
R1#
```

R1

```
reference time is E3695EA8.00000241 (22:52:24.577 UTC Tue Dec 29 2020)
clock offset is 0.00 msec, root delay is 1.00  msec
root dispersion is 56.97 msec, peer dispersion is 0.48 msec.
loopfilter state is 'CTRL' (Normal Controlled Loop), drift is - 0.000001193 s/s
system poll interval is 6, last update was 14 sec ago.
R1#show clock detail
2:52:46.320 GMT Wed Dec 30 2020
Time source is NTP
R1#
R1#
R1#
R1#
R1#
R1#
R1#
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#ntp master 8
R1(config)#ntp authenticate
R1(config)#ntp authenticate-key
R1(config)#ntp authenticate-key 1 md5 AnojLab
                                ^
% Invalid input detected at '^' marker.

R1(config)#ntp authentication-key 1 md5 AnojLab
R1(config)#ntp trusted-key 1
R1(config)#
```

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R2

```
R2>
R2>
R2>
R2>en
R2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#ntp autheticate
                    ^
% Invalid input detected at '^' marker.

R2(config)#ntp authenticate
R2(config)#ntp authentication-key 1 md5 AnojLab
R2(config)#ntp trusted-key 1
R2(config)#ntp server 192.168.12.1. key 1
                     ^
% Invalid input detected at '^' marker.

R2(config)#ntp server 192.168.12.1 key 1
R2(config)#
R2(config)#
R2(config)#
R2(config)#end
R2#
%SYS-5-CONFIG_I: Configured from console by console
```

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Top

R2

```
R2(config)#ntp autheticate
                    ^
% Invalid input detected at '^' marker.

R2(config)#ntp authenticate
R2(config)#ntp authentication-key 1 md5 AncjLab
R2(config)#ntp trusted-key 1
R2(config)#ntp server 192.168.12.1. key 1
                         ^
% Invalid input detected at '^' marker.

R2(config)#ntp server 192.168.12.1 key 1
R2(config)#
R2(config)#
R2(config)#
R2(config)#end
R2#
%SYS-5-CONFIG_I: Configured from console by console

R2#show ntp association

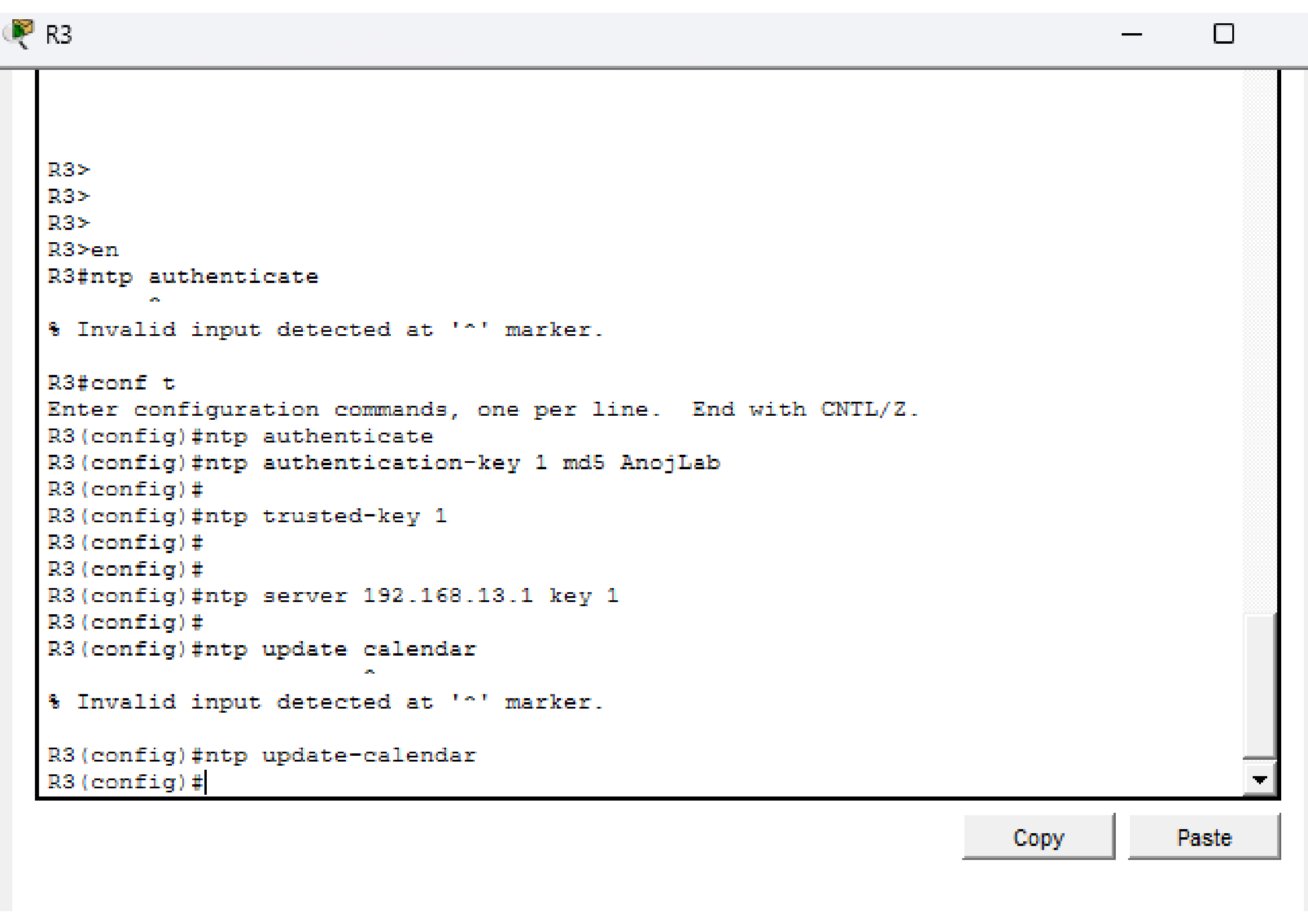
address         ref clock       st   when     poll    reach  delay           offset
disp
*~192.168.12.1  1.1.1.1         2    5        16      1      0.00            0.00
0.00
 * sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured
R2#
```

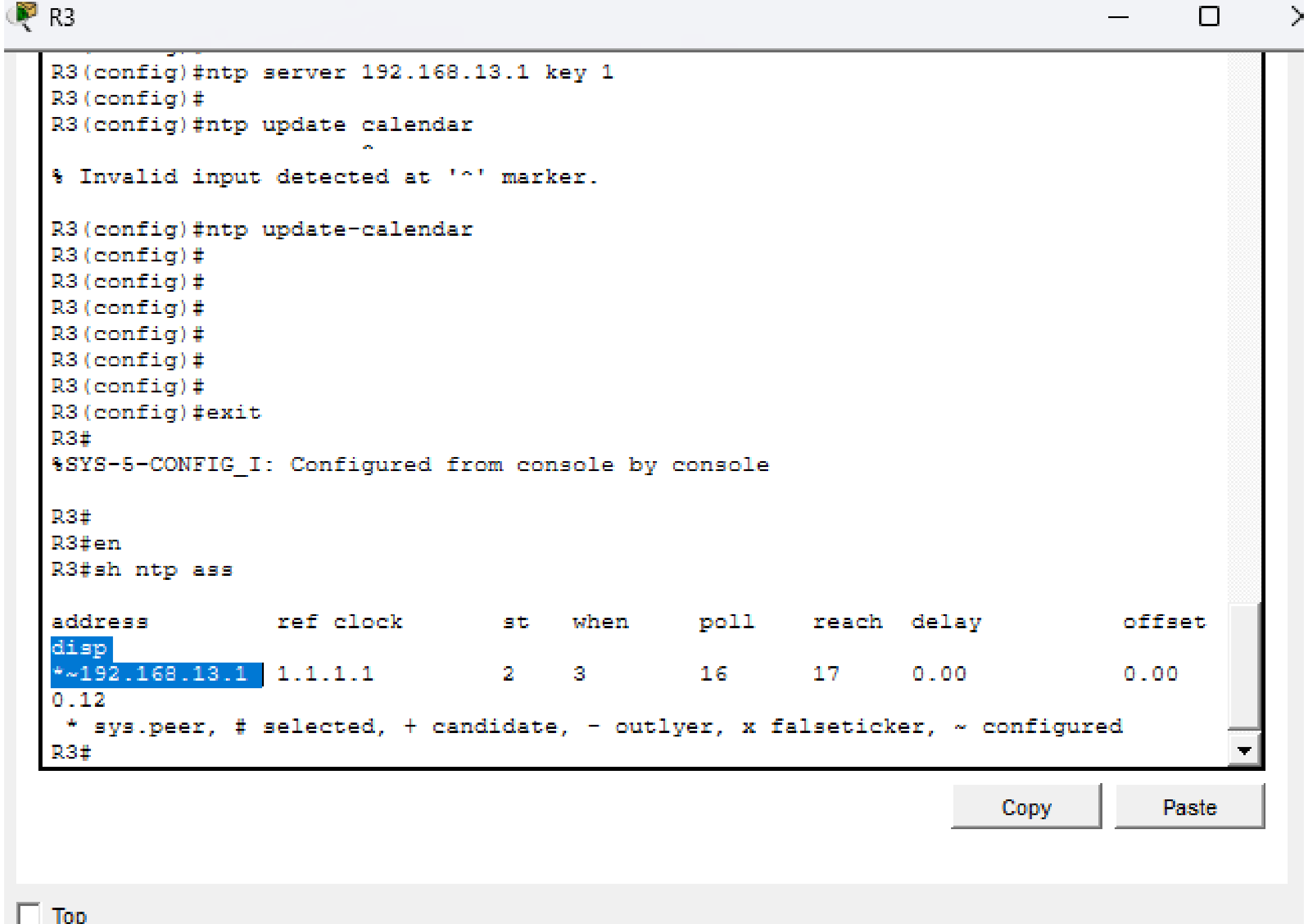
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Top

```
R2#sh clock detail
8:0:29.332 gmt Wed Dec 30 2020
Time source is NTP
R2#
R2#
R2#
R2#
R2#
R2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#ntp update calender
                            ^
% Invalid input detected at '^' marker.

R2(config)#ntp update-calendar
R2(config)#
```


R3
R3>
R3>
R3>
R3>en
R3#ntp authenticate
^
% Invalid input detected at '^' marker.
R3#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#ntp authenticate
R3(config)#ntp authentication-key 1 md5 AnojLab
R3(config)#
R3(config)#ntp trusted-key 1
R3(config)#
R3(config)#
R3(config)#ntp server 192.168.13.1 key 1
R3(config)#
R3(config)#ntp update calendar
^
% Invalid input detected at '^' marker.
R3(config)#ntp update-calendar
R3(config)#
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R3
R3(config)#ntp server 192.168.13.1 key 1
R3(config)#
R3(config)#ntp update calendar
^
% Invalid input detected at '^' marker.
R3(config)#ntp update-calendar
R3(config)#
R3(config)#
R3(config)#
R3(config)#
R3(config)#
R3(config)#
R3(config)#exit
R3#
%SYS-5-CONFIG_I: Configured from console by console
R3#
R3#en
R3#sh ntp ass
address ref clock st when poll reach delay offset
disp
*~192.168.13.1 1.1.1.1 2 3 16 17 0.00 0.00
0.12
* sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured
R3#
Copy
Paste
Top